

POSTER: VeilPIR: A Lightweight Private Information Retrieval Protocol for Enhancing Data Privacy in IoT Ecosystems

Saurav Ghosh
Southeast Missouri State University
Cape Girardeau, Missouri, USA
sghosh3s@semo.edu

Sanchari Das
George Mason University
Fairfax, Virginia, USA
sdas35@gmu.edu

Abstract

The rapid expansion of wireless IoT devices in domains such as smart homes, healthcare, and critical infrastructure has heightened concerns about data privacy, especially as many devices continue to access cloud services without encryption. Private Information Retrieval (PIR) protocols offer a way to protect query privacy by hiding the accessed record index, but most existing approaches rely on computationally intensive operations that exceed the capabilities of low-power IoT hardware. We present *VeilPIR*, a lightweight single-server PIR protocol designed specifically for resource-constrained devices connected over wireless links like BLE and Wi-Fi. *VeilPIR* shifts the heavy cryptographic workload to the server, allowing clients to send small encrypted queries and receive constant-size responses that preserve semantic privacy. We implemented *VeilPIR* in Python using Flask and deployed it to simulate realistic IoT conditions. Evaluation on a client and a cloud server shows 900 ms round-trip for 1,000 records and <1 J energy per query, while still allowing live record insertions and deletions without full re-encryption.

CCS Concepts

• **Security and privacy** → **Privacy-preserving protocols**; *Cloud computing security*; *Cryptography*; Mobile and wireless security; • **Computer systems organization** → *Cyber-physical systems*.

Keywords

PIR, IoT security, homomorphic encryption, data privacy

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1 Introduction

The Internet of Things now connects billions of small devices that monitor homes, hospitals, factories, and cities. These devices often run on minimal memory and battery power, yet they must exchange sensitive data with remote cloud servers. Standard cryptographic

tools, while strong, place a heavy load on such hardware and can quickly drain energy or exceed available bandwidth. As a result, many deployments still expose which database records a device reads, leaving user behavior open to profiling.

Private Information Retrieval protects query privacy, but most single-server designs require the client to encrypt a full selector vector as long as the database itself. That approach is too large for microcontrollers and too slow for real-time wireless links. Even recent optimizations trim bandwidth at the cost of extra computation that small sensors cannot handle. *VeilPIR* addresses these limits with a lightweight protocol tailored to resource-constrained IoT nodes. Instead of encrypting an entire vector, the client transmits only a few tiny ciphertexts, while the server performs the intensive homomorphic work and returns one constant-size response. This design keeps client computation and bandwidth almost constant, regardless of database size. The protocol also allows records to be added or removed without re-encrypting the whole dataset, so it adapts smoothly to live data streams. A reference implementation written in Python shows that *VeilPIR* can serve one-thousand-record databases with sub-second round-trip time and modest energy use. In our tests a 1,000-row lookup completes in about 900 ms while the client expends less than one joule, well inside a coin-cell budget.

This work makes four practical contributions. First, it presents a single-server PIR protocol that shifts the costly cryptographic steps to the cloud so the client performs only lightweight operations. Second, it offers a Python prototype that uses a simple one-hot encryption trick and still handles databases of the size commonly found in IoT settings. Third, it adds seamless insert and delete functions so the database can grow or shrink without any full re-encryption. Fourth, it delivers a detailed performance study that shows the protocol keeps encryption times short, power use modest, and server response times within the bounds needed for real-world wireless devices.

2 Related Work

3 Key Idea

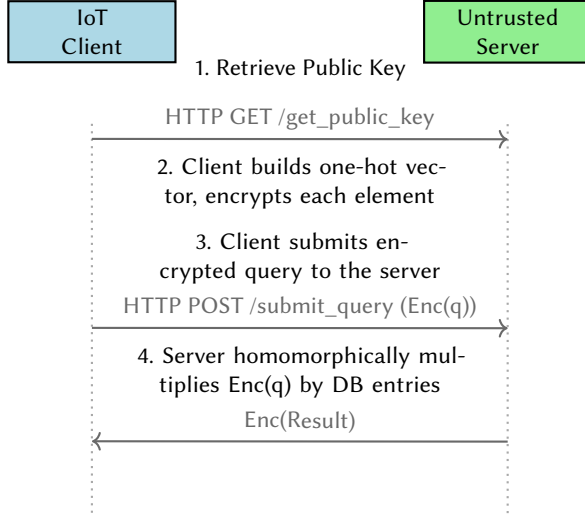
VeilPIR off-loads the expensive homomorphic work to the cloud while fixing client cost and response size. A sensor only encrypts a *one-hot* bit, so its CPU, memory, and uplink grow linearly with the database and remain well inside a coin-cell energy budget. The server sees ciphertexts only, yet a single addition over the homomorphic products yields the requested record. Dynamic inserts and deletes touch just one entry, avoiding global re-encryption and keeping latency stable as the dataset grows. See Figure 1 for the message flow.

^{*}The code implementation is available at <https://github.com/thetanmoy/VeilPIR>.

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Table 1: Evolution of Private-Information-Retrieval techniques and their IoT suitability

PIR family / era	Key work	Core idea	IoT impact
Multi-server IT-PIR (1995)	Chor et al. [5]	Query split among k non-colluding servers; information-theoretic privacy	Not fit for tiny sensors
Single-server C-PIR (1999)	Paillier [10]	Encrypts one-hot selector with additive HE; server computes on ciphertext	Cuts server count; uploads N -v
Compressed queries (2018)	SealPIR [3]	Uses FFT/batching to shrink bandwidth	Heavy on MCUs
Perf. trade-off studies (2021)	Ali et al. [2]	Empirical sweep over DB size, net speed, CPU	Reveals sweet spot
FHE-based PIR (2009–17)	Gentry [7], Cheon et al. [4]	FHE enables arbitrary ciphertext ops	Too slow / RAM-heavy
Sub-linear Server (2022–23)	Corrigan [6], Henzinger [8]	Index-aware transforms give $o(N)$ server cost	Exceeds our client budget
IoT security survey (2013)	Roman et al. [11]	Emphasizes ultra-low power	Inspires our minimal design

**Figure 1: UML Sequence Diagram for VeilPIR****Table 2: Latency and bandwidth per query.**

$ D $	Ciphertext KB	Enc. time (ms)	RTT (ms)
10	5.3	90	340
100	54	780	510
1 000	798	14 700	910

4 Prototype & Results

The codebase is small—about 450 lines of Python 3.9. A Flask server (150 loc) uses the phe Paillier library, while a 110-line client sends encrypted queries. Four REST endpoints cover key fetch, query, add, and delete. Records stay in RAM, or in SQLite when an env flag is set. After any insert or delete, the client only resizes its one-hot vector.

We ran six trials with 10 – 1 000 rows. Table 2 shows that, although encryption time rises with database size, round-trip latency stays under one second and the reply size is constant (3 KB).

Table 3 compares VeilPIR with three recent single-server PIR schemes. VeilPIR needs the least client effort and bandwidth, which makes it a practical choice for low-power IoT devices.

5 Conclusion

VeilPIR shows that tiny IoT gadgets can enjoy full query privacy without extra servers. A sensor sends one small ciphertext, the

Table 3: Single-server PIR schemes and IoT suitability (authors’ reading).

Scheme	Client work	Bandwidth	IoT ready
VeilPIR (this work)	low	constant	✓
SealPIR [3]	high	medium	✗
OnionPIR [9]	high	medium	✗
FastPIR [1]	medium	low	borderline

cloud does the hard math, and a reply comes back in under one second—even when the database holds 1 000 rows. The whole exchange uses less than a joule, so it fits within a coin-cell battery budget. Because record inserts and deletes never trigger a full re-encryption, the same design works for systems that grow over time, such as smart homes or factory sensors. In short, VeilPIR turns single-server PIR into a tool that everyday wireless devices can finally afford.

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